Algebra 1 Unit 7 Assessment Guide

General Information	Students learn to factor polynomials in unit 7. A student's ability to factor helps students solve quadratics and find a quadratic function's x-intercepts. This assessment includes a variety of polynomials. It is suggested that teachers consider student misconceptions based on these types of factoring: • Factoring out a common monomial • Factoring a trinomial where $a = 1$ • Factoring a trinomial where $a > 1$ • Factoring a difference of two squares • Factoring by grouping • Factoring using more than one technique This assessment intentionally does not present questions where $a < 0$. Students will have the opportunity to continue to build their skills and understanding of factoring in Unit 8, where students solve quadratics by factoring. As this unit is the first-time students learn factoring, ensuring that students have the six techniques listed in the bullet points will help students succeed in Unit 8. This assessment also deliberately limits the variables to x, y and z. To ensure that students are showing knowledge of the factoring techniques, the focus of the assessment is the process of factoring, not the preciseness of a variable. Although it is important for students to be able to be detailed and use the correct variable, this is a student's first assessment on factoring. As variable choice is often random, a student who mistakenly writes an incorrect variable may know the correct factoring technique and yet loses points on being precise with the variable. As a student becomes more fluent with factoring techniques, then the preciseness of variable can become an expectation.
Calculator Information	The questions on the unit assessment are calculator neutral. The teacher can determine whether to allow their students to use a calculator or not.
Total Recommended Points	The total recommended points in this guide are 100 points.

Question	1	Benchmark(s)	MA.912.AR.1.7
	$(5x+7y)(5x-7y)$	**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**2**	**Benchmark(s)**	MA.912.AR.1.7
	$(6x^2+5)(x-3)$	**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**3**	**Benchmark(s)**	MA.912.AR.1.7
	$6x^5y^4z^4(4y^2+9x^3z^5)$	**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.

Question	4	Benchmark(s)	MA.912.AR.1.7
$(4x+9)^2$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**5**	**Benchmark(s)**	MA.912.AR.1.7
$(x+2)(x-7)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**6**	**Benchmark(s)**	MA.912.AR.1.7
$(3x+2)(3x+5)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.

Question	7	Benchmark(s)	MA.912.AR.1.7
	$(2x + 9)(x - 4)$	**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**8**	**Benchmark(s)**	MA.912.AR.1.7
	$(x + 10)(x - 3)$	**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**9**	**Benchmark(s)**	MA.912.AR.1.7
	$(x^3 + 5)(x^3 - 5)$	**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.

Question	10	Benchmark(s)	MA.912.AR.1.7
$(x+9)(x+11)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**11**	**Benchmark(s)**	MA.912.AR.1.7
$(5x^2-1)(4x+1)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**12**	**Benchmark(s)**	MA.912.AR.1.7
$8y(3x^2-2x+7y^2)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.

Question	13	Benchmark(s)	MA.912.AR.1.7
$(x-7)(x-3)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**14**	**Benchmark(s)**	MA.912.AR.1.7
$8x^3y^2(1+7x^4y^3+4x^3y^2)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**15**	**Benchmark(s)**	MA.912.AR.1.7
$(x+8)^2$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.

Question	16	Benchmark(s)	MA.912.AR.1.7
$(4x-3)(2x+7)$		**Recommended Scoring** **(6 Total Points)**	Consider 4 points for student work and 2 points for the correct answer. Consider student work when determining partial credit. If a student makes a minor error that is carried through to the answer, consider whether to include a minor deduction of points.
Question	**17**	**Benchmark(s)**	MA.912.AR.1.7
☐ $2x(4x^2-6x-1)$ ☐ $2x^2(4x-6x-1)$ ☐ $4x^2(2x-3)-1(2x+3)$ ☐ $4x^2(2x-3)-1(2x-3)$ ☐ $(2x-3)(2x-1)(2x+1)$		**Recommended Scoring** **(4 Total Points)**	Consider 2 points for each correct answer. Consider deducting 1 point for each incorrect answer. Consider no points for selecting all answers.